

Refinement prompt (turn t)

Code-generation LLM
(open-weight model)

candidate revision (full files)

Security gate

1. Static analysis of the diff
(Semgrep, Bandit)

2. Graph-difference engine
 $\Delta G = G(t) - G(t-1)$, rules R1-R4

3. LLM triage of flagged deltas
(label and rationale)

Graph store
(NetworkX,
SQLite)

Decision: ALLOW or BLOCK

BLOCK: rationale added
to the next prompt

ALLOW: commit and
re-index changed files

re-index

